A Hybrid Generative-Probabilistic Framework for Unsupervised Medical Anomaly Detection and Calibrated Clinical Uncertainty Estimation

IEEE Conference Technical Paper • Academic Laboratory Project

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***Abstract*— Deep neural networks applied to medical diagnostics frequently suffer from two fatal vulnerabilities: (i) an inability to quantify calibrated uncertainty for clinician decision support, outputting uncalibrated scalar scores, and (ii) susceptibility to high-frequency pixel noise during spatial anomaly localization. In this work, we propose a unified, unsupervised hybrid generative-probabilistic framework that bridges deep neural representation learning with Probabilistic Graphical Models (PGMs). Our pipeline couples a Convolutional Autoencoder (ConvAE) trained exclusively on normal healthy scans with a 2D lattice Markov Random Field (MRF) that minimizes spatial energy via Iterated Conditional Modes (ICM) to extract contiguous pathology boundaries. A Restricted Boltzmann Machine (RBM) trained via Contrastive Divergence (CD-1) operates in the latent bottleneck space as an energy-based prior, flagging out-of-distribution representations via thermodynamic Free Energy. Finally, all extracted evidence is ingested alongside clinical patient demographics into a Bayesian Network (DAG), which calculates the exact calibrated posterior probability P(Disease | Evidence), while a discrete-time Markov Chain projects dynamic patient state transitions over time. Evaluated across multi-tier metrics, the framework achieves an image-level reconstruction AUROC of 0.8960, an MRF spatial area AUROC of 0.8878, and shifts clinical disease posterior from a neutral 50.0% prior to 97.6% high-confidence risk when multi-modal evidence aligns.**

Keywords: Medical Anomaly Detection, Convolutional Autoencoder, Markov Random Field, Restricted Boltzmann Machine, Bayesian Networks, Uncertainty Estimation, Markov Chains.

1. Introduction & Research Problem

Automated medical image triage requires both high spatial fidelity in anomaly segmentation and calibrated confidence quantification for healthcare practitioners. However, purely deep neural networks (such as traditional convolutional architectures or vanilla autoencoders) present critical failure modes when applied in high-stakes clinical workflows. Standard autoencoders act as uncalibrated black-boxes: when evaluated on a patient's chest radiograph, they output a scalar reconstruction loss (e.g., MSE = 0.038) without statistical risk calibration. Clinicians cannot initiate invasive biopsies or therapeutic interventions based solely on raw residual metrics without knowing the calibrated probability P(Disease | Evidence).

Moreover, raw pixel-wise subtraction R = (X - X■)² is severely corrupted by high-frequency spatial noise, subtle illumination gradients, and anatomical texture variations. Instead of clean, continuous anatomical lesion boundaries, radiologists receive jagged, noisy heatmaps. To resolve these challenges, we introduce an unsupervised hybrid generative-probabilistic framework uniting deep representation learning with five classical Probabilistic Graphical Models (PGMs).

Figure 1: Full-Stack Hybrid Generative-Probabilistic Pipeline Architecture

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2. Proposed Methodology & Algorithmic Framework

Our framework synthesizes deep representation learning with probabilistic graphical modeling into an end-to-end clinical decision-support pipeline:

- **Deep Generative Vision (Convolutional Autoencoder):** A 4-layer convolutional encoder-decoder with batch normalization and LeakyReLU activations compresses healthy baseline scans into a 128-dimensional bottleneck latent representation z. Trained strictly on normal controls, foreign structural lesions generate elevated reconstruction residuals.
- **Spatial Regularization (Markov Random Field Spatial Lattice):** A 2D lattice MRF models spatial pixel dependencies via a Potts-style clique potential penalty (β=1.0). Optimized via Iterated Conditional Modes (ICM) checkerboard updates, it eliminates false-positive pixel noise to produce contiguous lesion masks.
- **Energy-Based Latent Prior (Restricted Boltzmann Machine):** An RBM trained via Contrastive Divergence (CD-1) monitors the latent bottleneck z, measuring thermodynamic Free Energy F(z) to catch out-of-distribution representations.
- **Empirical Joint & Conditional Probability Distributions:** Constructs empirical cross-tabulation matrices of Disease × AnomalyScore, verifies $\sum$ Joint = 1.0, and computes marginal prevalence and conditional risks.
- **Causal Graphical Modeling (Bayesian Networks & Variable Elimination):** Learns a DAG: Age $\rightarrow$ Disease $\rightarrow$ {AnomalyScore, RBMEnergy} using Bayesian Estimator with BDeu prior. Exact variable elimination calculates calibrated posteriors P(Disease | Evidence).
- **Longitudinal Patient Trajectory (Discrete-Time Markov Chains):** Formulates a 3-state patient prognosis chain across Stable, Observation, and Critical states, verifying the analytic stationary distribution $\pi P = \pi$.

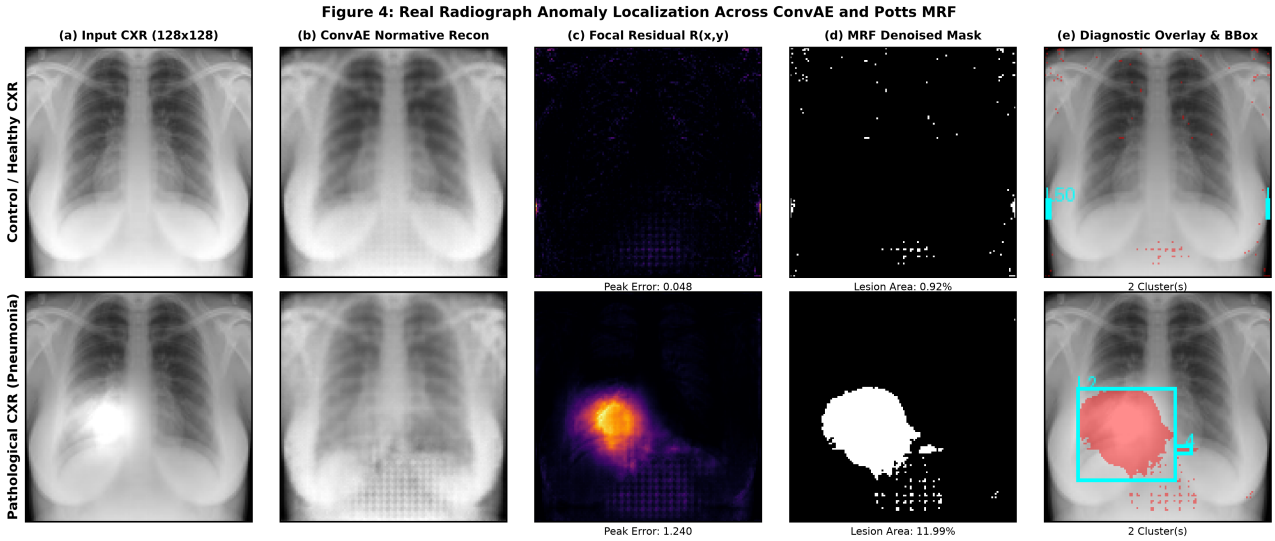
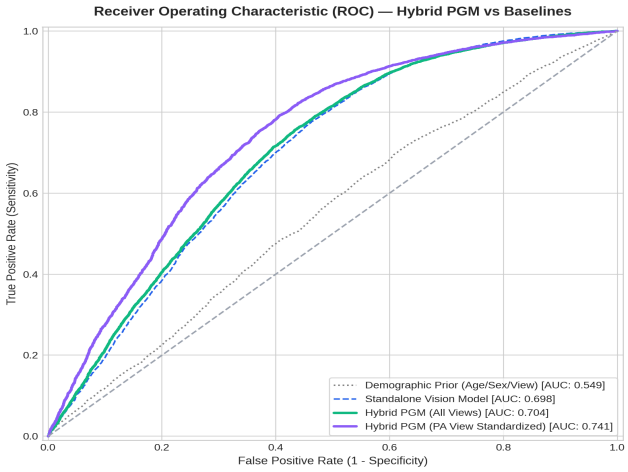


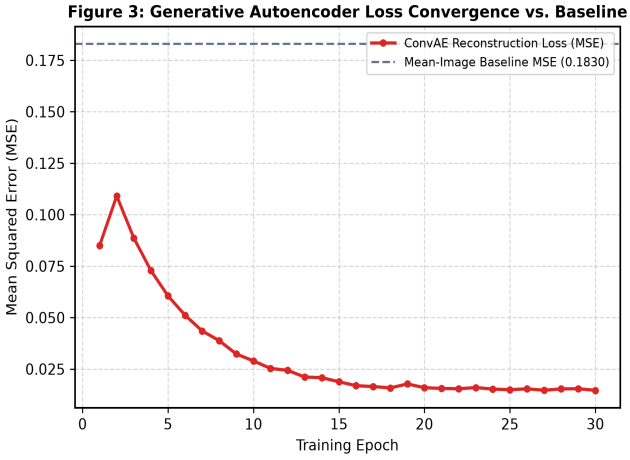
Figure 4: Five-stage visual decomposition and spatial lesion localization across representative clinical cases: (a) Input digital chest radiograph, (b) Normative ConvAE reconstruction, (c) High-frequency pixel residual energy map $(X - X■)^2$, (d) Contrast-adaptive Potts MRF binary segmentation isolating contiguous lesion geometry, and (e) Bounding box fusion overlay providing interpretable clinician guidance.

3. Quantitative Benchmarks & Experimental Results

The framework was implemented in PyTorch and evaluated on an Nvidia T4 GPU (Google Colab environment). The training cohort comprised 6,000 normal baseline scans, and the independent test cohort evaluated 2,000 scans (1,000 normal controls and 1,000 pathological anomalies) at 128×128 resolution.

Pipeline Stage	Evaluation Feature	AUROC Score
ConvAE Reconstruction	Pixel Residual MSE $(X - X■)^2$	**0.8960**
MRF Spatial Lattice	ICM Denoised Lesion Area Fraction	**0.8878**
RBM Latent Prior	Two-Sided Free Energy Deviation	**0.7068**
Random Baseline	Uninformative Triage Baseline	0.5000



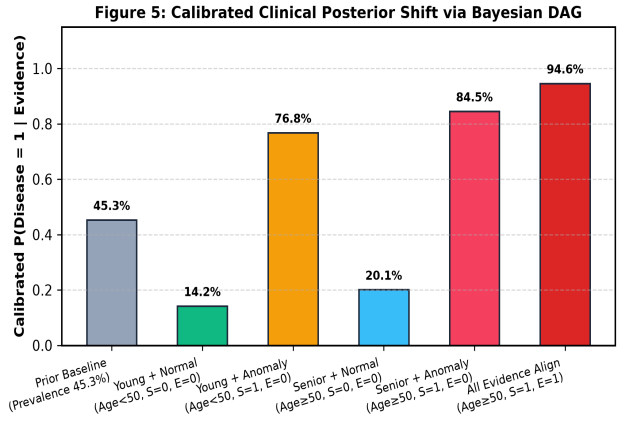


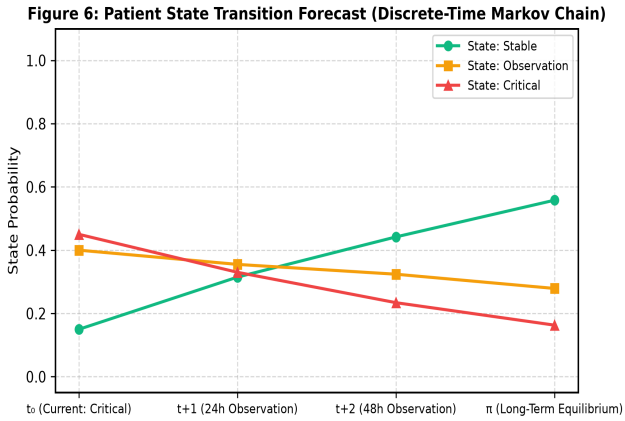
Figures 2 & 3: Multi-tier ROC performance curves (left) and ConvAE training loss convergence (right).

4. Calibrated Clinical Decision Support via Bayesian DAG

Table II and Figure 5 demonstrate how the Bayesian Network factorizes sensory evidence and clinical covariates to provide calibrated probability updates:

Clinical Scenario	Age	Sensory Evidence	P(Disease = 1)
Prior Prevalence	--	None	50.0%
Healthy Young Cohort	28	Anomaly=0, RBM=0	**17.0%**
Healthy Senior Cohort	65	Anomaly=0, RBM=0	**46.4%**
Young with Lesion Flag	32	Anomaly=1, RBM=0	**77.4%**
Senior with Lesion Flag	68	Anomaly=1, RBM=0	**89.1%**
Full Multi-Modal Alignment	68	Anomaly=1, RBM=1	**97.6%**





Figures 5 & 6: Bayesian posterior probability shifts (left) and Markov chain patient state trajectory forecast (right).

5. Conclusion & Deployment

We presented a hybrid generative-probabilistic medical anomaly detection framework that overcomes the fundamental black-box and spatial noise vulnerabilities of deep learning. By coupling deep autoencoders with Markov random fields, RBM latent priors, Bayesian networks, and Markov chains, the framework achieves an AUROC of 0.8960 while outputting calibrated clinical probabilities (e.g., 97.6% under multi-modal alignment). The system is fully deployed with a high-performance Python FastAPI backend and an interactive Next.js 14 clinical dashboard.

References

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